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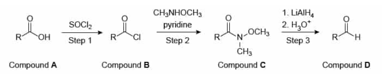
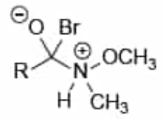


Scheme I

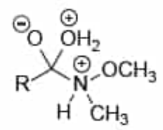
Which of the following is a structure of a reaction intermediate for Step 2 of the synthesis in the passage?

○ A.



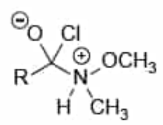
[2%]

○ B.



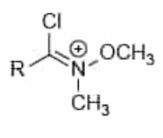
[11%]

✔ ◉ C.



[83%]

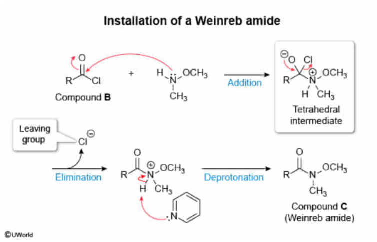
○ D.



[2%]

Correct
83% answered correctly

Explanation:



An **intermediate** is any substance that is formed and later consumed prior to the completion of the reaction. Consequently, an intermediate is a product of an earlier mechanistic step that becomes a reactant for a later mechanistic step.

Carboxylic acid derivatives differ in the group bonded to the carboxylic acid carbonyl and are interconverted through nucleophilic acyl substitution reactions. A nucleophilic acyl substitution is a type of addition-elimination mechanism. Initially, the reaction nucleophile attacks the electrophilic carbonyl carbon, shifting the pi electrons to a lone pair on the oxygen atom (addition). This step produces a **tetrahedral intermediate**, named for the sp^3 molecular geometry of the former carbonyl carbon. In the second step (elimination), the carbonyl pi bond reforms and a leaving group is ejected.

Step 2 of the synthesis in the passage is a nucleophilic acyl substitution between Compound **B** (an acid halide) and the Weinreb amine. Addition of the Weinreb amine to Compound **B** generates a tetrahedral intermediate containing a chlorine leaving group, which later collapses to produce Compound **C**. **Choice C** depicts the correct structure of the tetrahedral intermediate for Step 2 of the synthesis.

(Choice A) While this structure shows a tetrahedral intermediate, the leaving group is a *bromine* atom. Compound **B** is an *acyl chloride*, not an *acyl bromide*.

(Choice B) The leaving group in this structure is a molecule of neutral water, which requires a reaction between a strong acid and a hydroxyl group. Step 2 includes pyridine (a base), and Compound **B** does not contain a hydroxyl group.

(Choice D) This structure depicts the iminium ion formed if the Weinreb amine condensed with the carbonyl carbon of Compound **B** through a nucleophilic addition reaction. Nucleophilic additions are typically only observed for ketones and aldehydes, functional groups that lack a good leaving group.

Educational objective:
Carboxylic acid derivatives interconvert through nucleophilic acyl substitution reactions, which proceed through a tetrahedral intermediate. A tetrahedral intermediate contains an sp^3 carbon atom and is present just prior to the elimination of the leaving group.

Time spent:0 sec
Subject:
Organic Chemistry

QID:404031
System:
Functional Groups and Their Reactions

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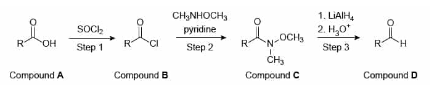
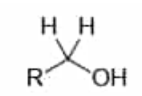
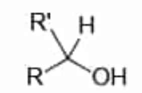
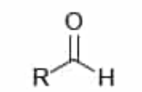


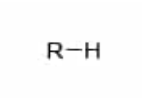
Scheme I

If Compound **A** replaced Compound **C** as the reactant in Step 3, what would be the structure of the product of this reaction?

A.  [59%]

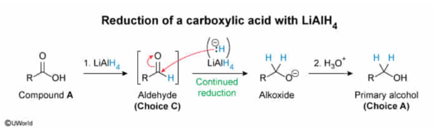
B.  [10%]

C.  [27%]

D.  [3%]

Correct
58% answered correctly

Explanation:



Oxidation-reduction (redox) reactions are an important class of transformations that involve the transfer of electrons between reactants. The processes of **oxidation** (loss of electrons) and **reduction** (gain of electrons) are inherently tied to one another and cannot occur in isolation. In the context of organic chemistry, an oxidation can be defined as

- the insertion of oxygen [O];
- the addition of molecular oxygen (O_2) or a molecular halogen (X_2); or
- the loss of molecular hydrogen (H_2).

By comparison, a reduction can be defined as the reverse of the above processes.

Lithium aluminum hydride ($LiAlH_4$) is a potent reducing agent that is a source of hydride ions. Hydride ions react as nucleophiles and donate their lone pairs of electrons to electron-deficient (electrophilic) molecules. $LiAlH_4$ reductions tend to use an excess of reagent, and it is generally challenging to use this reagent with controlled stoichiometry.

This question asks for the reaction outcome if Compound **A** (a carboxylic acid) is treated with $LiAlH_4$ followed by acidic workup. A strong reducing agent like $LiAlH_4$ will **fully reduce** carboxylic acids and most of their derivatives, adding **two equivalents** of hydride ion per molecule. Consequently, carboxylic acids will be initially reduced to a primary alkoxide. Acidic workup (H_3O^+) will protonate the alkoxide to a primary alcohol.

(Choice B) Given the additional presence of two R groups, this choice is a secondary alcohol. $LiAlH_4$ is not a source of the additional carbon atoms (R') for this type of transformation.

(Choice C) The *monoreduction* of a carboxylic acid by a hydride ion source will produce an aldehyde. Since $LiAlH_4$ is *more reactive* with an aldehyde than with a carboxylic acid, it is *not possible* to stop the reduction after only one hydride addition.

(Choice D) This structure has lost the entire carboxylic acid functional group through a decarboxylation reaction. Treatment with $LiAlH_4$ does not lead to the decarboxylation of carboxylic acids.

Educational objective:
Lithium aluminum hydride ($LiAlH_4$) is a strong reducing agent that delivers equivalents of hydride ion. When treated with $LiAlH_4$, carboxylic acids undergo a double hydride reduction to a primary alcohol.

Time spent:0 sec
Subject:
Organic Chemistry

QID:404032
System:
Functional Groups and Their Reactions

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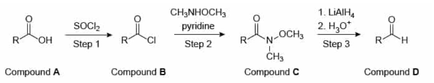


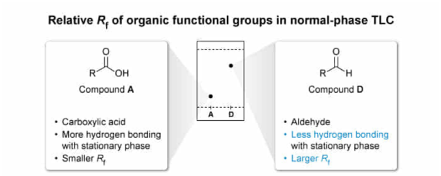
Scheme I

If Compounds **A** and **D** were analyzed by normal-phase thin-layer chromatography, which substance would have a larger R_f?

- A. Compound A, because it is more polar [33%]
- B. Compound A, because it is less polar [4%]
- C. Compound D, because it is more polar [4%]
- ✔ D. Compound D, because it is less polar [57%]

Correct
57% answered correctly

Explanation:



Aldehydes (Compound D) interact less with the polar stationary phase and have a **larger R_f**

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Thin-layer chromatography (TLC) is a diagnostic laboratory technique that separates substances based on differences in polarity. To perform TLC, a small quantity of material is applied to a plate containing a thin coating of stationary phase. The plate is placed in a mobile-phase reservoir and the solvent passes up the plate by capillary action.

In normal-phase TLC, the stationary phase is polar (eg, silica gel, alumina) and the mobile phase is nonpolar (eg, hexanes, ethyl acetate). Due to the principle of "like dissolves like", nonpolar compounds that more strongly interact with the nonpolar mobile phase will proceed closer to the solvent front. In contrast, polar compounds that more strongly interact with the polar stationary phase will not proceed as far. The relative mobility of a substance is quantified using the retention factor (R_f), which is calculated as the fractional percentage of mobility versus the solvent front. Consequently, in normal-phase TLC, **nonpolar compounds will have a larger R_f**, while polar compounds will have a smaller R_f.

In this question, Compound **A** contains a carboxylic acid group and Compound **D** contains an aldehyde group. While both functional groups contain a moderately polar carbon-oxygen double bond, as a carboxylic acid, Compound **A** has the capacity to hydrogen bond. As a result, Compound **D** is substantially **less polar** than Compound **A** and would be expected to have the **larger R_f**.

(Choices A and B) The capacity for hydrogen bonding makes Compound **A** *more polar* than Compound **D**. As a result, Compound **A** would be expected to have a *smaller* R_f in normal-phase TLC.

(Choice C) While Compound **D** will have a larger R_f in normal-phase TLC, this is because it is *less* polar than Compound **A**, not *more* polar.

Educational objective:
Normal-phase thin-layer chromatography separates compounds by polarity, where polar substances interact more strongly with the polar stationary phase. Carboxylic acids can hydrogen bond and are substantially more polar than aldehydes. Consequently, an aldehyde will have a larger retention factor than that of a carboxylic acid.

Time spent: 0 sec	**QID**:404033
Subject:	**System:**
Organic Chemistry	Functional Groups and Their Reactions

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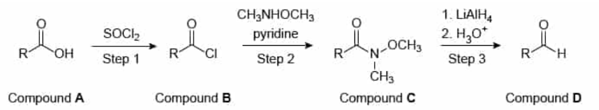


Scheme I

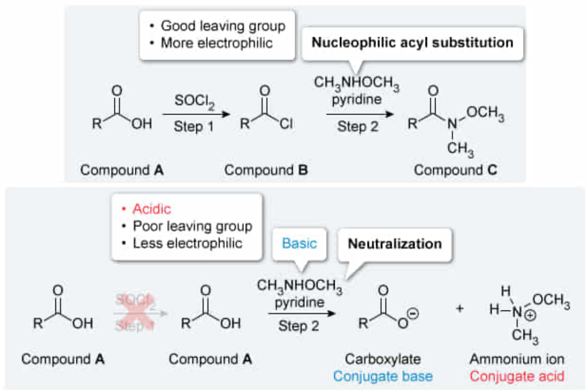
If Compound **A** is NOT subjected to Step 1 before proceeding to Step 2, what type of chemical transformation will occur in Step 2?

- A. Condensation [40%]
- B. Neutralization [19%]
- C. Elimination [18%]
- D. Hydrolysis [21%]

Correct
19% answered correctly

Explanation:

Impacts of eliminating a reaction step in a synthesis



Skipping Step 1 (activation of the carboxylic acid) turns Step 2 into a **neutralization reaction**.

Carboxylic acid derivatives differ in the group bonded to the carboxylic acid carbonyl and are interconverted through nucleophilic acyl substitution reactions. The **relative reactivity** of carboxylic acid derivatives results from a combination of steric effects and electronic effects. The general order of carboxylic acid derivative reactivity is, from most reactive to least: acid halide, acid anhydride, ester, amide, and carboxylic acid.

Viable carboxylic acid derivative interconversions are thermodynamically favorable when they proceed from **higher-reactivity derivatives to lower-reactivity derivatives**. For this reason, the conversion of a carboxylic acid to an acid halide (through reaction with $SOCl_2$) has a special role in synthesis design, as it transforms the lowest-reactivity derivative into the highest-reactivity derivative and facilitates most other acid interconversions.

This question asks what type of chemical reaction occurs if Compound **A** skips Step 1 and proceeds directly to Step 2. The condensation of a carboxylic acid and an amine into an amide (Compound **C**) is **not spontaneous**, as the carboxylic acid lacks a good leaving group and is less electrophilic than the acid halide (Compound **B**) that is formed during Step 1. Instead, Step 2 would promote a neutralization reaction between the carboxylic acid in Compound **A** and the basic reagents from Step 2: CH_3NHOCH_3 (an amine) or pyridine.

(Choice A) A condensation reaction is when two molecules are joined through a new covalent bond, often eliminating either a molecule of water or another small leaving group. Compound **A** lacks a good leaving group and is *unable* to directly condense through a reaction with the amine in Step 2.

(Choice C) An elimination reaction is when two groups are removed from adjacent carbons to form a new pi bond. Neither Compound **A** nor the contents of Step 2 contain a good leaving group to promote an elimination reaction.

(Choice D) A hydrolysis reaction occurs when a molecule of *water* promotes the cleavage of a bond. Step 2 does not include water.

Educational objective:

Interconversions of carboxylic acid derivatives generally proceed from derivatives of higher reactivity to lower reactivity. The direct reaction between a carboxylic acid and a basic amine will result in a neutralization reaction, rather than a condensation reaction.

Time spent:0 sec
Subject:
Organic Chemistry

QID:404034
System:
Functional Groups and Their Reactions